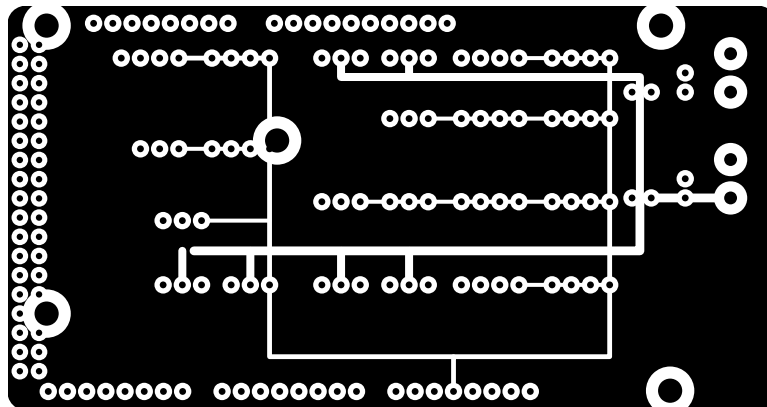
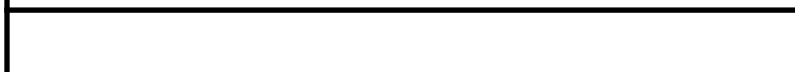




CALIBRATION CHECK BAR = EXACTLY 100.0 mm (Measure with physical caliper/ruler)



PECODROP RVM – ARDUINO MEGA 2560 EXPANSION SHIELD (REV 8.0 DIY ETCH MASK)

PRINT & FABRICATION INSTRUCTIONS (TONER TRANSFER / CHEMICAL ETCHING):

1. **LASER PRINTER SETUP:** Print on A4 Glossy Photo Paper or Laser Film at 100% Scale ('Actual Size').
CRITICAL: Do NOT select 'Fit to Page' or 'Shrink Oversized Pages' in Acrobat / Reader print dialog!
2. **VERIFY ACCURACY:** Check the 100.0 mm calibration bar above with digital calipers. It must measure 100.0mm.
3. **TONER TRANSFER:** Clean copper laminate with steel wool & acetone, iron at 200°C for 3 minutes, then soak.
4. **CHEMICAL ETCHING:** Etch in Ferric Chloride (FeCl3) or Sodium Persulfate at 45°C with agitation (8-12 mins).
5. **DRILL SPECIFICATIONS:**
 - Component Pads (Sensors, Servos, Mega Headers): 0.8 mm carbide bit
 - Power Screw Terminals (TB1, TB2) & Large Caps: 1.2 mm carbide bit
 - Chassis Standoff Mounting Holes (5x): 3.2 mm drill bit (M3 Screws)
6. **COMPATIBILITY:** 100% matched to PecoDropDesktopApp/Arduino/RVM_Arduino/RVM_Arduino.ino:
 Plastic Bin Full (Pin 47), Metal Bin Full (Pin 48), Paper Bin Full (Pin 49), MQ-6 Alarm (Pin 50).